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# Dual Process Theory for Large Language Models: An overview of using Psychology to address hallucination and reliability issues

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**Samuel C Bellini-Leite**

**Abstract**

State-of-the-art Large Language Models have recently exhibited extraordinary linguistic abilities which have surprisingly extended to reasoning. However, responses that are unreliable, false, or invented are still a frequent issue. It has been argued that scaling up strategies, as in increasing model size or hardware power, might not be enough to resolve the issue. Recent research has implemented Type 2 strategies (such as Chain-of-Thought and Tree-of-Thought), as strategies that mimic Type 2 reasoning, from Dual Process Theory, to interact with Large Language Models for improved results. The current paper reviews these strategies in light of the Predicting and Reflecting Framework for understanding Dual Process Theory and suggests what Psychology, drawing from research in executive functions, thinking disposition and creativity, can further contribute to possible implementations that address hallucination and reliability issues.

**Keywords**

Large Language Models, Chain-of-Thoughts, Tree-of-Thoughts, hallucinations, Dual Process Theory

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## 1. Introduction

Recent success in Artificial Intelligence (AI) has made it clear that some form of intelligence can in fact be accomplished by information processing. Since AI is reaching human competence in some benchmarks (OpenAI, 2023), a comparison between human cognitive models and AI models is becoming increasingly warranted (Pezzulo et al., 2023). By understanding mentality as information processing, if similar enough (Fodor, 1968), an AI model can be taken as a simulation of a psychological theory of cognition, and thus as a test of a cognitive model, and Psychology can suggest what is left for AI to accomplish.

Clark (2013b, 2016) argued human reasoning could be explained by predictive processing alone. Deep learning neural networks (Wang & Raj, 2017) bear various similarities to how Friston (2005, 2010) and Clark (2013b, 2016) describe cognition, as based on knowledge-rich predictive probabilistic generative models. The initial success of deep learning, in addition to the proposal for transformers (2017), culminating in Large Language Models (LLMs) and creative AIs, made it seem initially that Friston (2010) and Clark (2016) were correct in arguing that predictive probabilistic distributed processing in large neural networks is roughly all there is to cognition.

However, recurring issues have been occurring to AI based solely on probabilistic neural networks. In LLMs (OpenAI, 2023), these include hallucinations, basic mistakes in keeping track of reasoning or calculation steps, and lack of reliability, especially in long-range reasoning (see also, Valmeekam et al., 2022). We can see similar cases on image generators that represent an image incorrectly, for instance a hand with seven fingers. Apparently, this seems to be an intrinsic flaw of probabilistic neural networks as Wang and Raj (2017, p. 42) explain "Since neural networks are typically forced to make a prediction, it is not surprising to see a network classify a meaningless pattern into something […]."

If one is not careful with prompts or without applying further reasoning modules, when tested on human reasoning

Minas Gerais State University, Belo Horizonte, Brazil

**Corresponding author:**
Samuel C Bellini-Leite, Minas Gerais State University, R. Gonçalves Dias, 1434–Lourdes, Belo Horizonte 30140-092, Brazil.
Email: samuel.leite@uemg.br

tasks, GPT-3.5 usually fails in ways that are similar to human failures. For instance, it gives the intuitive but incorrect responses of the Cognitive Reflection Task (Frederick, 2005). Although state-of-the-art models such as GPT-4 were trained specifically in reasoning, OpenAI (2023)admits you can still find similar flaws if you use different tasks. In humans, these would be explained as Type 1 biases following Dual Process Theory.

Dual Process Theory (Evans & Stanovich, 2013; Kahneman, 2011) is an approach to human reasoning and decision-making that suggests two distinct types of processes are involved, an intuitive one (Type 1) which is faster, autonomous, and less effortful and an analytic one (Type 2) which loads working memory, is slower in comparison and may make use of decoupled representations (Stanovich & Toplak, 2012).

The Predicting and Reflecting Framework (PRF; Bellini-Leite, 2017, 2022; Bellini-Leite & Frankish, 2020) for DPT argues that although predictive processing adequately accounts for the largest part of human cognition, humans still need another fundamentally different type of process for explicit representations, compositional reasoning, and to explore alternative steps in reasoning. Following the PRF, these biases stemming from predictive architectures can be overridden by the executive functions of a working memory (Baddeley, 1992) executing heuristic search as proposed by Newell and Simon (1976) and Newell (1980).

Interestingly, recent papers (Hsieh et al., 2023; Shinn et al., 2023; Si et al., 2022; Wang et al., 2022; Wei et al., 2022; Yao et al., 2023) have been finding results of improvement on LLMs' benchmarks by simulating Type 2 processes[1] with classic symbolic AI. This paper reviews these strategies, discusses them in relation to the PRF, and based on human psychology, suggests possible new strategies for AI going forward.

## 2. The predicting and reflecting framework

The Predicting and Reflecting Framework for Dual Process Theory (Bellini-Leite, 2017, 2022; Bellini-Leite & Frankish, 2020) is a framework that combines predictive processing (Clark, 2016) with symbolic classical approaches (Newell, 1980) in the context of human reasoning, judgment, and decision-making (see Figure 1). It can be thought of as an update of Sloman's (1996) DPT according to current Psychology and Neuroscience. The hypothesis suggests that Type 1 (T1) processes, which involve intuitive and fast judgments, are based on predictive processing (Clark, 2016), while Type 2 (T2) processes, which involve deliberate reasoning, rely on symbolic classical cognition (Newell, 1980).

The hypothesis builds on Kahneman's (2002) suggestion of a connection between intuitive thinking and perception, noting that intuitive judgments are perception-like and involve the operations of "System 1." The hypothesis states that T1 processes, similar to perception, generate predictions as their output. These predictions are influenced by prior occurrences and current states and are shaped by "expectations" (Clark, 2013b). T2 processes then receive these T1 predictions as input for analysis and possible override.

According to the hypothesis, T1 processes deal with content encoded as probability density functions, which are distributed across brain regions and connected to external organs and body parts. These processes are subpersonal (see Frankish, 2009b) and do not represent discrete objects but rather refer to statistical aspects of the input. Expectations play a crucial role in T1 predictions, which can be biased by manipulating prior information. For instance, psychologists can use priming (McNamara, 2005) and framing (E Stanovich & West, 1998) (a prompt engineer can manipulate LLMs in a similar fashion).

On the other hand, T2 processes are described as classical reasoning machines, similar to Newell and Simon's (1963) General Problem Solver, designed to organize thinking. T2 processes operate within a predictive processing network that generates T1 responses. T2 processes investigate a problem space where an initial problem and a predicted solution, derived from T1 processing, are represented, by using move generators to reduce differences between the initial expression and the predicted expression, allowing for various problem-solving strategies related to heuristic search (Newell & Simon, 1976), and for tracking specific reasoning steps that were explored.

Symbolic representations are more easily grasped by subjects. They can easily verbalize and think about them, while probabilistic representations are fuzzier and harder to articulate. Classical cognitive architectures, using symbolic representations, may offer more fixed access to content than predictive processing networks. Sometimes there's a need for exact, unambiguous reference to some content, and Clark (2016) introduces the concept of single peak probability distributions which focus on one primary outcome, similar to a discrete symbolic representation. However, when this occurs, the advantages of predictive processing's flexibility is lost, but there is a gain in certainty. Clark's proposition that density functions can occasionally have one primary value essentially eliminates ambiguation and makes the content more accessible, in effect, it acts as a symbolic representation. Traditional symbols are easily graspable because they can be stored and manipulated in working memory. Drawing from Fodor and Pylyshyn (1988), symbolic representations allow for compositionality in cognitive processes, making them more trackable, since a constituent in one expression is the same in other expression it appears, allowing for step-by-step reasoning. For the PRF, this difference in representation format is fundamental to explaining what is unique in each process.

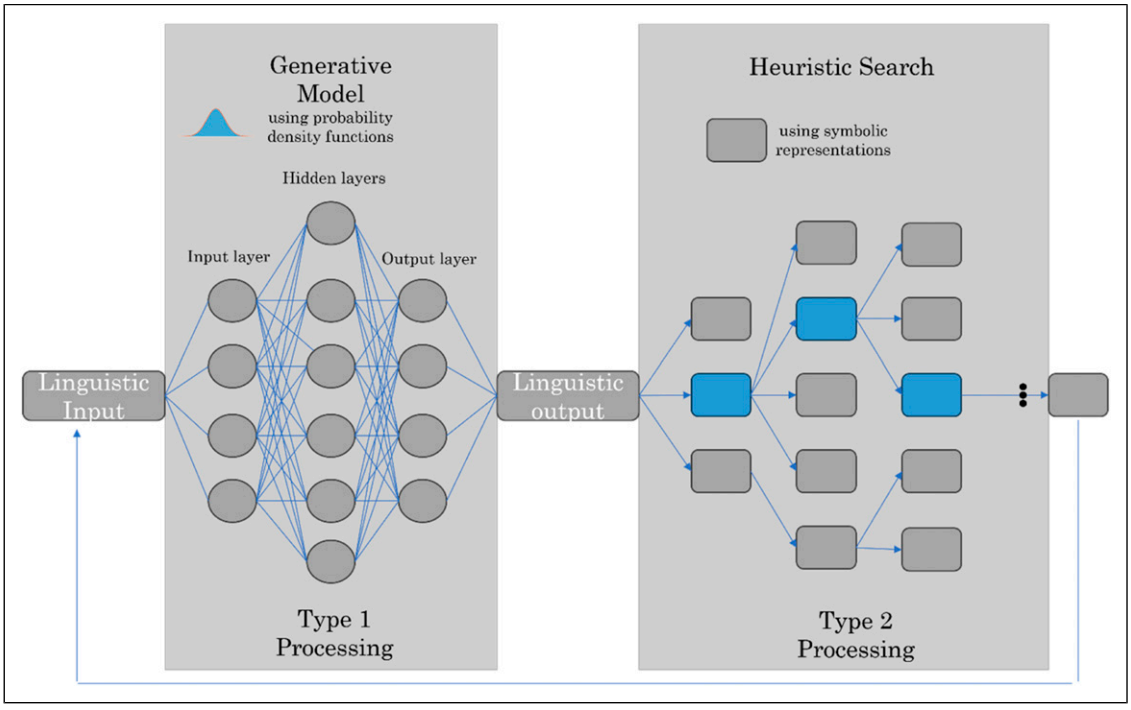


**Figure 1.** Visual depiction of the predicting and reflecting framework which can be representative both of human reasoning and of tree-of-thoughts applied to large language models.

Further, predictive processing contrasts with working memory, a component of cognitive psychology that is key to effortful thinking. Essentially, when predictive processing is not effective (high prediction error, high free energy), is when working memory is called in. While literature on predictive processing tends to sideline working memory, it remains a central tenet in Cognitive Psychology. Notably, a classical architecture—akin to a Von Neumann structure or a Turing machine—has a storage mechanism and processes that mediate symbolic processing, which are pivotal in the functioning of working memory. The way Turing Machines function, as described by Alan Turing, closely mirrors the properties of working memory.

Although a similar PRF structure has recently been proposed in another modality (Ding, et al., 2023), the hypothesis is limited to reasoning, judgment, and decision-making and does not propose dual processes for the entire mind (Bellini-Leite, 2018).

It is important to note that PRF is a development of DPT and DPT itself has received criticism (Keren, 2013; Keren & Schul, 2009; Kruglanski, 2013; Kruglanski & Gigerenzer, 2011; Mercier & Sperber, 2017; Osman, 2004, 2013). DPT on its own does not explain how these two types of processes work in algorithmic terms, in the sense of Marr's (1982) three levels of description. Exploiting vulnerabilities, critics often argue for a single system of reasoning (Keren & Schul, 2009), or that predictive processing networks can account for DPT through different levels of processing in the same architecture (Clark, 2016). The PRF goes against such criticism by developing a framework for the algorithmic level of explanation for DPT that includes a solution to the unity problem (Samuel, 2009), that is, it explains what makes each type of process computationally unique.

Also, there are some differences to the received interpretation of DPT, in which T1 processes are seen as "dumb," or "blind," quick and dirty mechanisms and general-purpose reasoning itself is left for T2 processes to accomplish. This is an issue because T2 processes under a classical computational framework having to accomplish general reasoning results in computational explosion or the frame problem (Pylyshyn, 1987). Having T1 as a predictive generative model changes what is expected of T2 as well, since it then searches over contextualized generative model outputs rather than requiring some form of idealized general rationality (Bellini-Leite & Frankish, 2020).

In what follows, we review how current progress in AI is showing that the DPT structure in general is useful, and in particular that the PRF, by showing the differences between predictive processing and classical computing in DPT, beyond solving unity problem (Samuel, 2009), has been predictive of new successful strategies to mitigating hallucination and reliability issues in AI (Hsieh et al., 2023; Shinn et al., 2023; Si et al., 2022; Wang et al., 2022; Wei et al., 2022; Yao et al., 2023).

## 3. Addressing the hallucination and reliability limitations of LLMs

According to OpenAI (2023), GPT-4, the state-of-the-art LLM, demonstrates notable improvements compared to earlier versions. However, it still exhibits certain limitations that need to be carefully considered. GPT-4 exhibits a significant reduction in hallucinations and reasoning errors compared to previous models, as confirmed by internal, adversarial factuality evaluations. It achieves a 19% improvement over GPT-3.5, demonstrating progress in accurately separating fact from incorrect statements.

OpenAI (2023) report goes on to explain that, despite advancements, GPT-4 still faces challenges related to accuracy, reasoning, and contextual understanding, and that it may still make simple reasoning errors inconsistent with its broad competence. The model can also be gullible in accepting false statements and occasionally fails to solve complex problems. GPT-4 displays calibrated predictions in its pre-trained state, aligning predicted confidence with the probability of correctness. However, after post-training, the model's calibration is reduced. This implies that GPT-4 may not double-check its work when the likelihood of errors is high, potentially leading to confidently incorrect answers.

OpenAI (2023) also acknowledges the presence of biases in GPT-4's outputs and are still making efforts to mitigate it. As Wang and Raj (2017) note, these are all expected issues of a deep leaning neural network. As Bellini-Leite (2017, p. 60) suggested from the PRF: "In fact, a hypothesis we can extract from this is that predictive processing systems would be subject to bias from lack of compositionality, such as mistakes in transitivity, failures in noticing necessary character of formal rules and so on […]." This has recently been confirmed of transformers (Dziri et al., 2023) and LLMs (Berglund et al., 2023), and it is not clear that making LLMs more powerful in the sense of adding more data or powerful hardware will address these gaps (Rae et al., 2021).

However, users have noted that adequately prompting LLMs can have a powerful effect on making their answers more coherent and avoiding these abovementioned gaps. Following this perception, an increasing trend (Hsieh et al., 2023; Shinn et al., 2023; Si et al., 2022; Wang et al., 2022; Wei et al., 2022; Yao et al., 2023) has been emerging in AI to design prompting strategies to be coupled onto LLMs to organize prompting or reflect on responses, without the need for human intervention. After all, if the user needs information from the LLM, they are likely not experts in the topic, so they cannot orient it adequately. This trend led to what we can call *Type 2 strategies*, which are strategies that generally aim to automize the T2 process that human prompters are applying to the output of LLMs. In what follows, these strategies will be reviewed, and argued for considering the PRF.

### 3.1. Chain-of-thought

Wei et al. (2022) explored a novel approach to unlock the reasoning abilities of LLMs harnessing the potential of in-context few-shot learning through prompting. Chain-of-Thought (CoT) prompting involves providing the model with a prompt consisting of triples: input, chain of thought, and output. The CoT represents a series of intermediate natural language reasoning steps that lead to the final output.

For LLMs, the ability to tackle complex reasoning tasks, such as multi-step math word problems, is a challenging feat. Humans usually decompose such problems into intermediate steps, solving each one before arriving at the final answer. CoT prompting endows LLMs with a similar capability by generating coherent chains of thoughts, which mimic step-by-step reasoning processes (Hsieh et al., 2023).

Nye et al. (2021) demonstrated how a scratchpad, that is, allowing for demonstration of intermediate reasoning steps, enhanced Transformers' proficiency in performing long addition. This enhancement led to better out-of-distribution generalization for more extensive problem instances and aided in polynomial evaluation.

This approach allows for the allocation of additional computational resources to address problems that necessitate more reasoning steps. By breaking down complex tasks, models can effectively navigate through the intricacies and arrive at accurate solutions. This is in line with what is expected of T2 reasoning in humans (Evans & Over, 1996).

Chain-of-Thought serves as an interpretable window into the behavior of language models. It provides insights into how a model may have arrived at a particular answer and offers opportunities for debugging (Chen, Schärli & Zhou, 2023) when the reasoning path goes astray. This is explained in the PRF (Bellini-Leite, 2017) as the ability of classic systems to have compositionality (Fodor & Pylyshyn, 1988) and explicit representations to pinpoint reasoning steps overcoming the probabilistic nature of representations in predictive networks (For similar points see Nye, 2020)

According to Wei et al. (2022), one of the significant advantages of CoT reasoning is its ease of elicitation in large, off-the-shelf language models. By including examples of CoT sequences in the exemplars of few-shot prompting, models can readily learn and generate coherent CoT. This approach harnesses the power of existing language models, enabling them to exhibit sophisticated reasoning abilities without significant modifications.

Wei et al. (2022) show effectiveness of CoT prompting through empirical evaluations on arithmetic, commonsense, and symbolic reasoning. Using just eight chain-of-thought exemplars, a PaLM 540B model achieved previous state-of-the-art accuracy on the GSM8K benchmark of math word problems (Cobbe, et al., 2021), surpassing a finetuned GPT-3 with a verifier. The PaLM 540B model, when prompted

with chain-of-thought, outperformed standard prompting by a large margin on the GSM8K benchmark of math word problems. The gains from chain-of-thought prompting were more pronounced for more complicated problems. For instance, on the GSM8K dataset, performance more than doubled for the largest GPT and PaLM models.

Similarly, Suzgun et al. (2022) found that applying CoT prompting to hard big bench tasks (BBH; Srivastava et al., 2022) enabled the PaLM model to surpass the average human-rater performance on 10 out of the 23 tasks. Codex (code-davinci-002) surpassed the average human-rater performance on 17 of the 23 tasks when using CoT prompting. Suzgun et al. (2022) go on to present a comparison of performance using standard "answer-only" prompting and CoT prompting on the BBH tasks. PaLM 540 B with answer-only prompting outperformed the average human-rater on 6 out of 23 BBH tasks. With CoT prompting, this improved to 10 out of 23 tasks. InstructGPT (text-davinci-002) with answer-only prompting outperformed the average human-rater on 4 out of 23 tasks. With CoT prompting, this improved to 15 out of 23 tasks. Codex (code-davinci-002) with answer-only prompting outperformed the average human-rater on 5 out of 23 tasks. With CoT prompting, this improved to 17 out of 23 tasks.

### 3.2. Self-consistency

Wang et al. (2022) introduce a novel decoding strategy called self-consistency (CoT-SC) as a replacement for the conventional greedy decoding strategy utilized in CoT prompting. The self-consistency decoding approach significantly enhances language models' reasoning performance by capitalizing on the observation that complex reasoning tasks often admit multiple paths to reach the correct answer.

The CoT-SC decoding strategy involves two key steps. Initially, the language model is prompted using CoT prompting. However, instead of selecting the optimal reasoning path greedily, Wang et al. (2022) propose a "sample-and-marginalize" decoding procedure. This procedure entails sampling from the language model's decoder to generate a varied set of reasoning paths. Each path may lead to a different final answer. To determine the optimal answer, they marginalize out the sampled reasoning paths and identify the most consistent answer from the resulting set.

The CoT-SC approach draws an analogy to human reasoning experiences. When multiple different ways of thinking converge on the same answer, humans gain greater confidence in the correctness of the final answer. Similarly, by considering diverse reasoning paths and finding the most consistent answer among them, the self-consistency decoding strategy emulates this aspect of human reasoning (Xiong et al., 2022).

Wang et al. (2022) demonstrate that even when a fixed answer is desired, incorporating diversity in the reasoning processes can offer substantial advantages. By embracing self-consistency, this strategy explores a broader space of potential responses, leading to improved performance and a more nuanced understanding of the task at hand.

The concept of self-consistency deviates from traditional ensemble approaches, which rely on training multiple models and aggregating their outputs. Instead, it functions akin to a "self-ensemble" technique that operates on a single language model. By leveraging the inherent capabilities of pre-trained language models, self-consistency provides an effective and off-the-shelf solution for improving the coherence and consistency of generated text (see also Fluri et al., 2023).

Self-consistency showed improvements over chain-of-thought prompting across various tasks. Specifically, when combined with models like PaLM-540B or GPT-3, self-consistency achieved new state-of-the-art results on: GSM8K: +17.9% absolute accuracy gains; SVAMP (Patel et al., 2021): +11.0%; AQuA (Ling et al., 2017): +12.2%; StrategyQA (Geva et al., 2021): +6.4% and ARC-challenge (Clark et al., 2018): +3.9%.

Self-Consistency differs from society of mind approaches (Du et al., 2023; Zhuge et al., 2023), that have been recently proposed, which involves the utilization of multiple language model instances to collaboratively generate a unified answer to a given query. In this society of mind approach, each model instance initially generates its own candidate answers. Subsequently, all instances review and evaluate the responses provided by their counterparts, incorporating this feedback to refine their individual answers. This iterative process of reviewing and updating is repeated over several rounds, ultimately leading to a consolidated and improved answer. By leveraging the collective intelligence and reasoning capabilities of multiple language models, the approach enhances the accuracy and coherence of the final response. Such society of mind approach is more akin to the solution that Clark (2016) suggests for achieving higher-level reason, contrary to the PRF which proposes a classical computational procedure to explore alternatives on one LLM.

### 3.3. Reflexion

Shinn et al. (2023) propose an approach to help LLMs learn when interacting with external environments (through APIs, for instance) called Reflexion, a framework to reinforce LLMs through linguistic feedback instead of updating weights. Reflexion converts binary or scalar feedback from the environment into textual summaries, which are incorporated to the next learning episode. This self-reflective feedback serves as a semantic gradient signal emulating the

iterative learning process employed by human analytic reasoning when learning from previous mistakes.

Reflexion incorporates three modules: the Actor, the Evaluator, and the Self-Reflection model. The Actor component of the Reflexion framework is based on a LLM, it samples actions or generations from the current policy at each time step, receiving observations from the environment in return. Different strategies for Actor models, such as the CoT (Wei et al., 2022) and ReAct (Yao et al., 2022a), can be further incorporated.

The Evaluator takes a generated trajectory as input and computes a reward score that reflects the performance of the trajectory within the given task context. It determines when self-reflection should occur based on the current state, action, hyperparameters, and trajectory history. It considers the number of repetitive action cycles and the maximum number of actions allowed in an environment. The goal is to detect hallucination and inefficient planning (see also Valmeekam et al., 2022; Wang et al., 2023). Therefore, it relates to the overriding capacity of humans in DPT, which is mediated by individual differences, but how this is accomplished is still debated (Evans & Frankish, 2009).

In particular, it is debated whether T1 or T2 processes are involved in the detection of a need to override. PRF, following predictive processing, proposes that out of all processes occurring in the brain the processes relative to errors that receive higher precision-weighting (see Clark, 2013a) are the ones which will have T2 intervention. In this sense, deciding if further processing is needed is not itself a T2 process, but rather stems from the T1 predictive architecture. In the analogy, the LLM would need to decide when to call for reflection. However, on the PRF this need can also be driven through heuristic search exploration performed by T2 processes, since following unexpected lines of reasoning can generate prediction errors on the T1 generative models. It is an open research question whether this predictive processing mechanism for determining the need for T2 processes can draw from the Reflexion evaluator or vice-versa.

The Self-Reflection model, is the part of the Reflexion framework that attempts to apply the changes. By generating verbal self-reflections based on sparse rewards and persistent memory, it provides feedback for future trials. Unlike scalar rewards, this nuanced feedback offers specific insights into the agent's actions and their consequences. For instance, when faced with a failure signal in a multi-step task, the agent can analyze its previous actions and identify the specific action that led to subsequent incorrect choices (see Liu et al., 2023). By verbally acknowledging the alternative action it should have taken and storing this experience in its memory, the agent can adapt its decision-making approach in subsequent trials. This iterative process of trial, error, self-reflection, and memory persistence enables the agent to rapidly improve its decision-making capabilities across different environments, leveraging informative feedback signals for optimal performance (see also Madaan et al., 2023).

On the PRF, a symbolic identification of separate thought-instances is what gives the reasoning process the ability to track its reasoning steps, whereas in probability density functions used in T1, functions can vary, losing the meaning and the places in which it played a role previously. The self-reflection model could be seen as a possible way in which humans keep track of their reasoning steps. The memory component of the Reflexion framework is also crucial, as related to human short-term storages components of working memory (Baddeley, 1992). Further, within the Reflexion framework, the trajectory history serves as the short-term memory, providing immediate context for the agent's actions. On the other hand, the outputs generated by the Self-Reflection model are stored in the long-term memory, enabling the agent to learn from past experiences and incorporate valuable lessons into its decision-making process.

Reflexion achieved a 91% pass@1 accuracy, surpassing GPT-4, which achieved 80% on HumanEval Coding Benchmark (Chen et al., 2021). Reflexion agents improve on decision-making AlfWorld (Shridhar et al., 2020) over baseline approaches by 22% in 12 iterative learning steps. Reflexion improves on reasoning questions in HotPotQA (Yang et al., 2018) by 20% outperforming baseline approaches. Specifically, Reflexion helps the agent correct its mistakes without access to the ground truth answer, improving its accuracy by 14%. However, Reflexion faced challenges with the WebShop benchmark (Yao et al., 2022), which tests agents to navigate an e-commerce website. The agent did not show signs of improvement after four trials, indicating that Reflexion struggles with tasks requiring significant diversity and exploration.

The "society of minds" approach, utilizing multiagent debate, consistently outperforms the CoT and Reflexion in tasks including factual accuracy of biographies, multimodal language understanding, chess playing, and arithmetic tasks. However, these are not precisely the sorts of reasoning tasks Reflexion was working on. Neither Du et al., (2023) nor Zhuge et al., (2023) showed results on HumanEval, AlfWorld, or HotPotQA for a straightforward comparison.

### 3.4. Tree-of-thoughts

The Tree-of-Thoughts (ToT) proposed by Yao et al. (2023) is the first to implement a heuristic search system to reason on the output of LLMs, thus most adequately representing the PRF for human cognition. ToT is also inspired by DPT (Evans & Stanovich, 2013; Kahneman, 2011), and proposes the construction of a tree-like structure, where each node represents a partial solution or a state in the problem-solving

process. The tree is built by generating coherent language sequences called "thoughts" that serve as intermediate steps toward problem-solving. These thoughts are evaluated and explored using a deliberative reasoning process, similar to T2 thinking in human cognition (Evans & Over, 1996).

Depending on the nature of the problem, the intermediate thought steps can be decomposed into smaller or larger units, such as phrases, equations, or paragraphs. The thought generator generates potential thoughts from each state in the tree. The state evaluator assesses the progress made by each state in solving the problem. Two evaluation strategies are suggested: (a) evaluating each state independently based on value prompts, or (b) comparing and voting for the most promising state among a set of states.

The experiment presented by Yao et al. (2023) evaluates different prompting methods for the Game of 24 task, where the goal is to use four numbers and basic arithmetic operations to obtain a result of 24. The baseline methods include Input-Output (IO), CoT and CoT-SC prompting compared to the trees of thoughts (ToT). IO, CoT, and CoT-SC prompting methods perform badly on the task, achieving only 7.3%, 4.0%, and 9.0% success rates. In contrast, ToT with a breadth of b = 1 already achieves a success rate of 45%, while b = 5 achieves 74%.

Tree-of-Thoughts was also tested on a creative task. The assignment required the creation of a cohesive text comprising four paragraphs, each concluding with a specified random sentence. ToT produced passages that were more logically connected than those from both IO and CoT. When assessed by humans, passages generated by ToT were favored over CoT in 41 out of 100 instances, whereas CoT was chosen over ToT in just 21 instances (Yao et al., 2023).

In a 5 × 5 mini crosswords challenge, while IO achieved a 14% word-level success and CoT reached 15.6%, ToT notably excelled with a 60% word-level success, solving up to 7 out of 20 games, showing intricate reasoning and strategy planning (Yao et al., 2023).

Likewise, a similar ToT strategy was applied to sudoku (Long, 2023). This version of the ToT was compared to a Zero-shot solver (zs), which directly inputs the puzzle to the LLM; the One-shot solver (os), which supplements the puzzle with a step-by-step solution of Sudoku; and the Few-shot solver (fs), which offers the LLM multiple CoT styled examples. The ToT solver exhibited the highest efficacy, successfully surpassing other solvers.

Reflexion, ToT, and the Society of mind approaches are state-of-the-art additions to LLMs, and as of yet there has not been enough coherent testing to determine which is most efficient for what. The PRF predicts that, although the society of mind approach should be helpful and complementary (given that any sort of recursive re-checking is beneficial), there cannot be non-hallucinatory, reliable human-like reasoning without a T2 strategy for AI that incorporates a similar ToT along with elements that simulate a full-blown human-like working memory (somewhat in the direction of Reflexion), with storage and executive functions, autonomously prompting the generative model for information, processing on symbolic sentence representations (formed from the output of predictive generative models) that work with constituents and syntactical relations exhibiting compositionality and systematicity (Bellini-Leite, 2017, 2022; Fodor & Pylyshyn, 1988; Newell, 1980), since this is it what allows humans to override misleading T1 predictions and maintain a coherent and true story of state of affairs.

## 4. Further T2 reasoning abilities to be incorporated over LLMs

Since AI works by discovering algorithms that solve cognitive tasks, it is more likely that Psychology will benefit from AI when it comes to discovering *how* cognition works. However, Psychology can also offer scientific insights on what cognition does. Often in AI work, researchers use their intuition on what human cognitive abilities are. In what follows, I attempt to suggest what sort of T2 related processes the psychological literature has discussed that could still be incorporated into LLMs outputs to possibly improve the results.

### *4.1. Executive functions*

The AI strategies we have been reviewing can be seen as attempts to implement something like a working memory. Baddeley's (1992) model of working memory consists of storage components and executive functions, with two storage components specialized for maintaining speech-based and visual-spatial information, known as the phonological loop and visuospatial sketchpad, respectively (a third storage, the episodic buffer was also added later). While the phonological loop is very similar to a CoT structure, the visuospatial sketchpad could be an image simulator that is used as a prompt on an LLM (Kosslyn, 1996). Considering GPT-4 (OpenAI, 2023) has visual recognition and visual reasoning (Yang et al., 2023), this does not seem farfetched to implement (see also Hu et al., 2023; Wu et al., 2023; Zhang, et al., 2023b).

Perhaps what is most relevant for the analogy with LLMs is the central executive, the model's control structure, responsible for regulating cognitive processes. Although Psychology has not figured out executive functions entirely, the Miyake et al. (2000) model as well as Baggetta and Alexander (2016), suggest three very much agreed upon functions of executive control: shifting, updating, and inhibition.

Shifting, also known as attention switching or task switching, involves the ability to move between multiple tasks, operations, or mental sets. Miyake et al. (2000) explain shifting as the disengagement of an irrelevant task set and the subsequent engagement of a relevant task set. It might also involve overcoming interference or negative priming, especially when a new operation needs to be performed on a set of stimuli that had previously been subjected to a different operation.

Updating (Miyake et al., 2000) involves the monitoring and modification of working memory representations. Updating requires actively monitoring of incoming information, determining the relevance to the task at hand, and replacing old content with newer, more relevant content. It may involve temporal tagging to differentiate between old and irrelevant content. Updating goes beyond simple maintenance and involves dynamic manipulation of working memory contents.

Interestingly, recent research (DeCaro et al., 2016) has shown how having lower working memory capacity, thus disengaging from previous items, can help avoid fixation (Finke et al., 1996) and help in insight problem-solving. The same happens when we are tired of using working memory, a condition known as ego-depletion (Decaro, Van Stockum, 2018). Fixation in humans (Jansson & Smith, 1991; Smith & Blankenship, 1991) seems to be similar to the snowballing of hallucinations recently noted in LLMs (Zhang, Press, Merrill et al., 2023). Given this, some purposeful disengagement mechanism could be set in place for adequate shifting and updating when relevant (Shipstead et al., 2016).

The concept of inhibition involves the deliberate suppression of dominant or automatic responses when necessary. The inhibition performed by working memory refers to deliberate, controlled suppression of prepotent responses, and it does not encompass other forms of inhibition found in spreading activation models or connectionist networks. Although ToT and other T2 strategies are external to the LLMs internal processing and weight shifting, perhaps inhibition can come in the form of prompts that inhibit certain types of responses. For instance, working memory inhibition can play an important role in downplaying superstitious beliefs (Lindeman et al., 2011; Narmashiri et al., 2023; Svedholm & Lindeman, 2013), and could analogously play a role in downplaying hallucinatory content in LLMs.

Planning is also related to working memory. When reviewing the evidence, Gilhooly (2004) explains that in the Tower of London task (TOL, see Phillips et al., 2001), both older and younger participants were observed to follow a problem-reduction approach termed the "goal-selection strategy." This strategy involves identifying subgoals that can be quickly achieved. Specifically, participants tend to target discs that can be placed in their desired positions with minimal moves of obstructing discs to a holding peg. After addressing the most immediate subgoal, they proceed to the next easiest subgoal and continue in this manner. This approach results in very narrow search patterns, as observed in the study. The goal-selection strategy is characterized by its focus on achieving immediate subgoals that require the fewest moves, thereby streamlining the planning process. This method of tackling the most straightforward tasks first and progressively addressing more complex ones is indicative of a heuristic approach to problem-solving in the TOL task.

Finally, in the case of general behavior control and organization, which are usually thought of as executive functions, although we can think of the environment "prompting" humans, there is certainly some degree of internal "self-prompting." For instance, on Baddeley's (1992) version the working memory can prompt long-term memory for content. Working memory should also be what organizes our "self-prompts," making us autonomous agents. Interestingly, applications (Auto-GPT, 2023; Fernando et al., 2023; Zhang et al., 2022) have been attempting to replicate this sort of autonomy.

### 4.2. Thinking dispositions

Our T2 capacities supposedly play a significant role in shaping our flat-out beliefs, attitudes, and decision-making processes (Frankish, 2009a). Psychologists have identified several key thinking dispositions that reflect variations in people's goal management, epistemic values, and epistemic self-regulation (Toplak et al., 2014). These dispositions encompass a range of cognitive propensities, such as the tendency to gather information before reaching conclusions, the inclination to consider different perspectives, the inclination to think deeply about problems, and the ability to weigh evidence in relation to one's opinions. Furthermore, individuals differ in their inclination to contemplate future consequences, evaluate pros and cons before making decisions, and seek nuance rather than adhering to absolutist thinking; these differences affect human rationality (Stanovich et al., 2016).

Actively open-minded thinking ((Stanovich & Toplak, 2023; Stanovich & West, 1997) involves actively seeking information and considering various viewpoints before forming beliefs. Need for cognition (Cacioppo & Petty, 1982) reflects a disposition to engage in thorough thinking and analysis when faced with a problem, the disposition a person has to effectively engage in T2 thinking when needed. Consideration of future consequences (Strathman et al., 1994) entails thinking about potential outcomes of actions before taking them. Need for closure (Webster & Kruglanski, 1994) represents a desire for definitive answers and a reluctance to tolerate ambiguity. Finally, lack of

dogmatism (Shearman & Levine, 2006) represents a disposition to avoid rigid adherence to one's beliefs without considering alternative perspectives.

State-of-the-art LLMs are quite proficient at giving rational and adequate responses to prompts. However, the reliability issue still may make some responses fail in this regard. It would make sense to have built-in prompts that enforce thinking dispositions in the style of CoT. These could be done by using linguistics prompts based on thinking dispositions scales. In this regard it would be interesting to look at the Actively Open-Minded Thinking (Stanovich & West, 1998) and the others mentioned above; it could be fruitful for an AI T2 strategy to enforce actively open-minded thinking prompts, or as tools (Paranjape et al., 2023), on the LLM. For instance, "Beliefs should always be revised in response to new information or evidence."

Further, having the CoT strategy to successively raise awareness to common cognitive biases and content incoherences could be important. Studies have found that training interventions that made participants aware of their biases reduced the manifestation of those biases in subsequent decisions in a variety of settings (AlKhars et al., 2019; Croskerry, 2003; Morewedge et al., 2015; see also Willingham, 2008). Likewise, LLMs should use belief-like sentences that foster thinking dispositions through metacognition and be remembered to avoid biases. This could go along with verification processes (Dhuliawala et al., 2023; Ling et al., 2023; Zhao et al., 2023)

### 4.3. Creativity research

The Geneplore model (Finke et al., 1996) proposes a two-phase process for creative thinking: generation and exploration. In the generative phase, individuals construct mental representations called preinventive structures (see Finke, 1990), which serve as internal precursors to creative products. These structures possess properties that facilitate creative discovery. In the subsequent exploratory phase, the preinventive structures are interpreted and explored to derive meaningful insights. The cyclic nature of the model allows for iterations between the phases, enabling individuals to refine and expand their conceptual understanding. Successful explorations may lead directly to creative products, while unsuccessful attempts prompt either the generation of new preinventive structures or the modification of existing ones. On the PRF we can think of these two phases as the interaction of predictive T1 and reflective T2 processes, where T1 generates some possibilities and T2 investigates the generated state spaces to find interesting branches.

Creativity is characterized by the dynamic interaction between generation and exploration in the Geneplore model. Individuals engage in the generative phase by combining mental images, leading to visually intriguing forms. These forms are then interpreted and explored, often revealing the need for modifications or additional combinations. This iterative process allows for the generation of improved designs or unexpected interpretations. While repetitions of the cycle are not always necessary, they often facilitate creative discoveries. The model also considers the influence of constraints, which can be imposed during either the generative or exploratory phase, shaping the underlying cognitive processes. By acknowledging the diverse ways in which individuals excel in generating or interpreting preinventive structures, the model accounts for variations in creative style. It can be applied across different stages of creative thinking, from early idea generation to later problem-solving or design completion, offering a flexible framework to understand and foster creativity in various contexts (see Jaarsveld & Van Leeuwen, 2005; Purcell & Gero, 1998).

If we think how creativity is being explored in LLMs or image generation AIs, it seems to have more of a generative character than an exploratory one. Usually, the exploration is left for the user to accomplish, by choosing options, rearranging prompts, and by other human means. However, novelty in itself is not creative (Boden, 2007), we can generate a new sentence that is novel but is irrelevant. Creativity involves the development of some domain of knowledge, art, science and so forth, it includes finding something of value inside of the novel options that are generated. According to the PRF, to accomplish this, humans use T2 exploratory processes based on the output of predictive generative models.

According to Finke et al. (1996) creativity involves several exploratory processes that aid in the investigation of new ideas and possibilities. These processes include attribute finding, where emergent features are systematically sought in mental images or conceptual combinations. Conceptual interpretation which involves finding abstract or metaphorical meanings in preinventive structures, while functional inference explores the potential uses or functions of these structures. Contextual shifting involves considering preinventive structures in new contexts to gain insights into alternative uses or meanings. Hypothesis testing treats preinventive structures as possible solutions to problems and explores their implications (see also, Evans, 2007). Searching for limitations helps identify ideas that will not work and narrows the focus of creative exploration. Cropley (2006) also points out the role of convergent thinking to avoid risks, considering the limits of the project, seeking simplicity, and assessing feasibility.

These exploratory processes are typically carried out in an organized and systematic manner, with deliberate and controlled forms of processing predominating. They can be applied in various contexts, such as scientific theory development, design improvement, or artistic interpretation. Creativity involves clusters of these exploratory processes,

with different combinations being essential for different accomplishments (Cropley, 2006; Finke et al., 1996; Welling, 2007).

Users of AI image generators often iteratively refine prompts to yield more desirable images, effectively exploiting the model's potential through a trial-and-error process. Drawing inspiration from the Geneplore model, which posits that creativity encompasses both generative (coming up with ideas) and exploratory (refining and assessing those ideas) systems, one can envision an automated system that mimics this iterative refinement. By implementing a ToT agent to reason over the result of the first prompt, it could then suggest more refining strategies based on exploratory processes like attribute finding, hypothesis testing and searching for limitations (Evans, 2007; Finke et al., 1996). This is a promising strategy to try along with the new visual recognition features of LLMs (OpenAI, 2023; Yang et al., 2023). Likewise, this could be accomplished though mental models and graphs (Besta et al., 2023; Cao, 2023; Jiang et al., 2023; Johnson-Laird, 1983; Mumford et al., 2012). Over successive iterations, this process would ideally converge towards an image that optimally meets the desired criteria.

## 5. Limitations

It is common in the public debate to talk about the possibility of an Artificial General Intelligence (AGI) which includes all possible forms of reasoning. This paper, however, is focused solely on a small portion of human reasoning, which along with DPT, is focused on syllogism, analytic reasoning, decision-making under uncertainty, linguistic insight riddles and so on. Likewise, although LLMs already accomplish much of what we expect of reasoning, they still do not encompass all that is possible of human reasoning. For instance, we do not have LLMs that find new mathematical proofs, or single-handedly program a complex video game or propose functional new sciences. Clearly, although the development of these T2 strategies should help, they likely will not be enough to enable these capacities. If they are possible at all, multiple advancements should occur on both our understanding and applications of T1 and T2 processing, and maybe even other type of processes Psychology is not aware of. Further, the models that have already applied the T2 strategies we have been suggesting have only achieved mild improvements on LLM benchmarks, rather than being sufficient for a complete equivalent of human reasoning.

A recurring issue in this chain-of-thought literature is that different tasks and benchmarks have been used to assess LLM performance with these different strategies, so the comparison of success can still be somewhat arbitrary. However, since we do have adequate available benchmarks for testing (Zheng et al., 2023), this is mainly an issue of organizing the literature, which hopefully this overview can contribute with. Further, experimental papers should be concerned with specific benchmarks that follow directly from what they attempt to implement in their work, rather than any sort of improvement over GPT-4.

Although predictive processing was presented as functioning like an AI generative model, there are differences which have been described as challenges for AI to overcome (Pezzulo et al., 2023). Active inference is proposed to underscore the intricate relationship between predictions, actions, and the state of affairs in the world. The behavior of living organisms is rooted in grounded world models that generate predictions about the consequences of actions. For instance, when an organism contemplates moving in a particular direction, such as north or south, it forms predictions about the potential outcomes of that action, like the location it might end up in. These predictions actively influence the organism's understanding of hidden states in the world, such as its location in a spatial context. In essence, the act of predicting is intertwined with action, shaping the organism's interaction with its environment and its understanding of the world's state of affairs.

Considering the PRF uses predictive processing as the framework for explaining T1 processes, of course this also has implications for the general structure described in this paper. Predictive processing in the brain showcases a hierarchical structure with distinct "descending" and "ascending" pathways. The former subtracts predictions to compute errors, while the latter uses errors to update the model in real time action. Further, generative models in the brain are not receiving passive linguistic inputs but rather attempt to predict sensory states. Thus, human reasoning would be better represented in Figure 1 if the generative model included details such as ascending and descending flow of information and predictions were made upon each new step of the reasoning process, instead of the final output of the heuristic search going back to feed the model through a new input as is the case of the current T2 AI strategies.

Embodiment, which has been related to predictive processing (Clark, 2015) was not addressed. From the perspective of this paper, the challenges of embodiment can be seen as the challenges engineers are facing in transferring LLM intelligence over to the real world in robots (Open X-Embodiment Collaboration, 2023; Taniguchi et al., 2023). As we have seen, the Webshop task (Yao et al., 2022) was already an issue for Reflexion. Clearly, various new forms of reasoning would be required in the case of embodiment, such as spatial reasoning, reasoning regarding movement, affordance selection, and a general mapping of linguistic reasoning to real world reasoning. However, it is not entirely clear that we cannot have a completely virtual abstract reasoning machine nonetheless. In the same way Pezzullo et al. (2023) argue that embodiment is fundamentally

necessary to achieve meaning, Johnson and Lakoff (2002) had previously argued that embodiment is fundamentally essential for language, but we now have disembodied agents that excel in linguistic abilities.

## 6. Conclusions

It might be speculated that human working memory or T2 reasoning processes are more knowledge-rich and have more contextual capacities than these AI T2 strategies. However, it has previously been argued (Bellini-Leite & Frankish, 2020) that these knowledge and contextual capacities of the human brain must stem from predictive networks, since we have learned that classical serial symbolic computation is slow and inefficient to deal with context (Pylyshyn, 1987). Therefore, it is likely the case that the main role of working memory and T2 processes is to organize "prompts," to the brain's predictive network in a very similar way that these reviewed systems are organizing prompts on LLMs. In this picture, the organized and contextual cognition that we see healthy (but not necessarily brain damaged) humans exhibit is likely the result of an emergent subtle balance between these reflective and predictive algorithms. However, how this is precisely accomplished is a matter of figuring out technical details on AI, and the suggestions in section 4 are some of the clues Psychology has to offer.

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### ORCID iD

Samuel C Bellini-Leite https://orcid.org/0000-0003-0178-0267

### Note

1. Sometimes called a "system 2"—but see Bellini-Leite, 2018 for a review of issues with this terminology.

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## About the Author

Samuel C Bellini-Leite Psychologist, PhD in Philosophy—Logic, Science, Mind, and Language—at the Federal University of Minas Gerais (UFMG), focusing on various areas related to Cognitive Science: Cognitive Psychology, Philosophy of Psychology, Educational Psychology, Cognitive Software Design, Creativity and Cognition, Philosophy of Technology and Artificial Intelligence, Philosophy of Mind, and Theoretical Cognitive Science. Currently teaches at the Minas Gerais State University.